Deven Desai

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SUMMARY

Software Engineering Master's student and Stanford Hackathon winner specializing in scalable backend infrastructure, low latency data processing, and full stack web applications. Demonstrated track record at Terminal Trends and Parspec engineering microservices in Node.js and Python, reducing API response times by 40 percent, and building high precision document processing pipelines.

PROJECTS

AgenticFlow | Concurrent Distributed Microservices Engine GitHub

- Automated database transformations and complex schema migrations across cloud systems by architecting an independent microservices engine using Python, Go, and gRPC.
- Increased service execution throughput by 40 percent by designing concurrent execution loops, lock free communication patterns, and low overhead microservice interfaces.
- Maintained strict data consistency and transaction privacy across system execution boundaries by engineering an automated differential privacy validation layer.

Low Latency Search Engine | High Performance Analytical Processing Platform GitHub

- Analyzed and queried high volume structured datasets in real time by engineering a custom full stack analytical search interface as a solo developer using Java and React.
- Decreased search query response latency by 35 percent by implementing custom indexing algorithms and memory structures as measured by end to end benchmark traces.
- Increased processing throughput by 20 percent by tuning underlying data structures and batch data execution routines.

OmniTrack AI | Full Stack Real Time Data Processing Platform GitHub

- Processed unstructured streaming inputs and user metadata natively in cloud infrastructure by engineering a zero to one full stack application using Python and React.
- Achieved sub 100ms vector retrieval latency by independently optimizing Qdrant indexing algorithms for compute intensive, concurrent requests.
- Enhanced system stability and output correctness by 30 percent by building continuous execution monitoring and automated verification test suites.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin Concepts, Go, C++, TypeScript, JavaScript, SQL (Postgres)
Backend & Financial Systems: Microservices Architecture, RESTful APIs, gRPC, Distributed Systems, High Concurrency Pipelines
Full Stack & Web: React, Redux, Node.js, Express, HTML5 CSS3, State Management, API Design
Databases & Infrastructure: PostgreSQL, MySQL, Redis, Relational Schema Design, Docker, AWS Concepts, Git

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.